Modularity, Nestedness, and Resilience in an Urban Ecological Interaction Network

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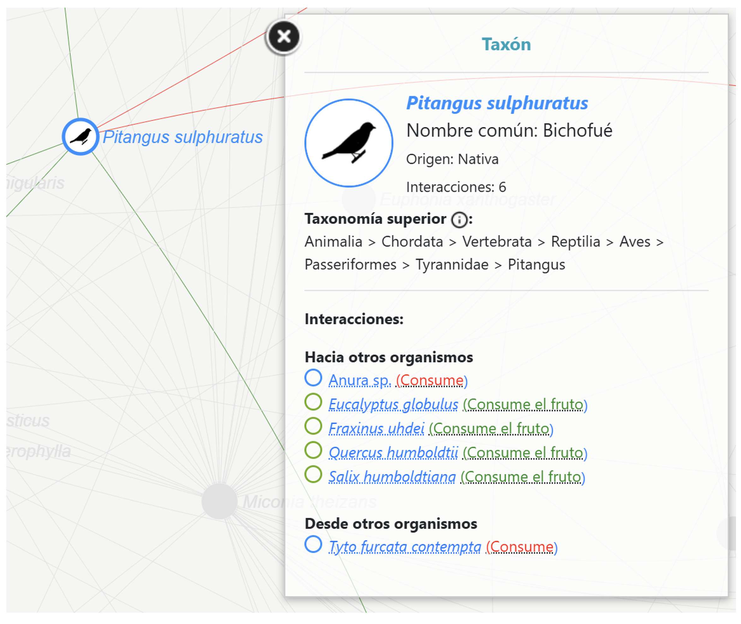
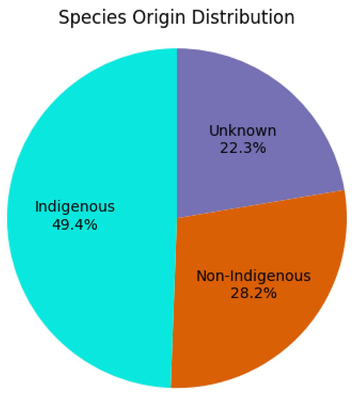
1 Abstract

A biotic interaction network of Bogotá was analyzed using centrality and architectural metrics to compare the roles of indigenous and non-indigenous species and to evaluate the network's resilience to node removal. Nestedness (NODF) and modularity (greedy Newman algorithm) were computed across mutualistic and parasitic inter-kingdom subnetworks. Native and non-native species showed no significant differences in standard centrality distributions, although the highest-ranked nodes were mostly non-native. Simulated species removal revealed a marked sensitivity to the elimination of highly connected nodes. These results suggest that non-native species tend to occupy structurally influential positions, often associated with non-mutualistic interactions, and that the network's stability is strongly dependent on a small set of highly connected species.

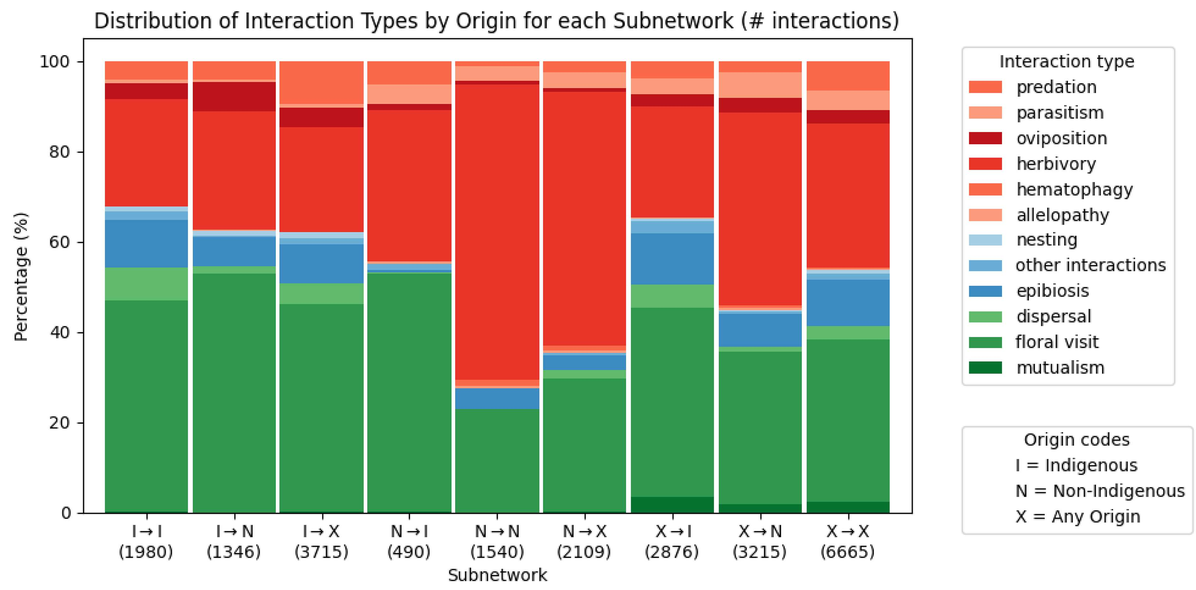
2 The Network [1]



Build from observed interactions between species of multiple kingdoms and origins



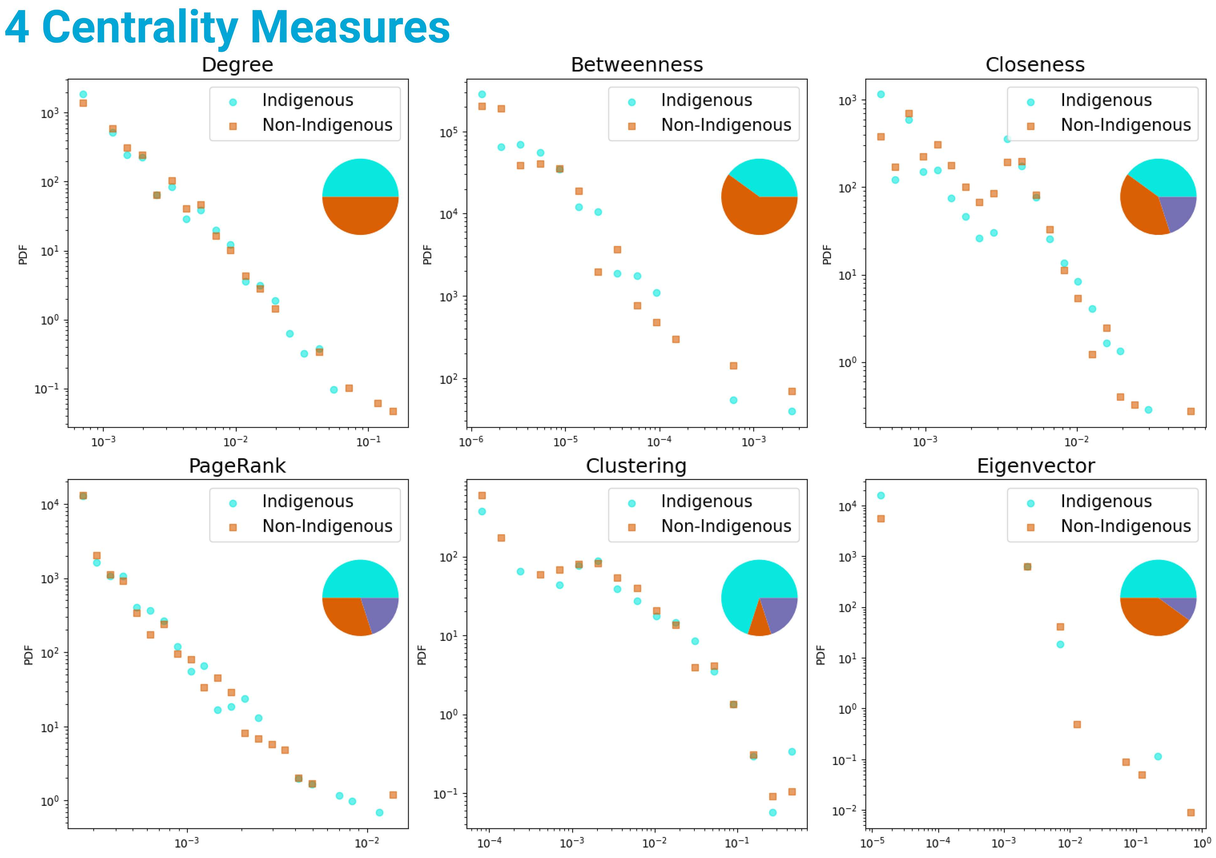
3 Interaction Type Distributions



Subnetworks based on species origin were constructed

Indigenous species tend to have more mutualistic interactions

4 Centrality Measures



No significant differences in network centrality distributions

Top 10 piecharts show a tendency for higher ranked species to be Non-Indigenous

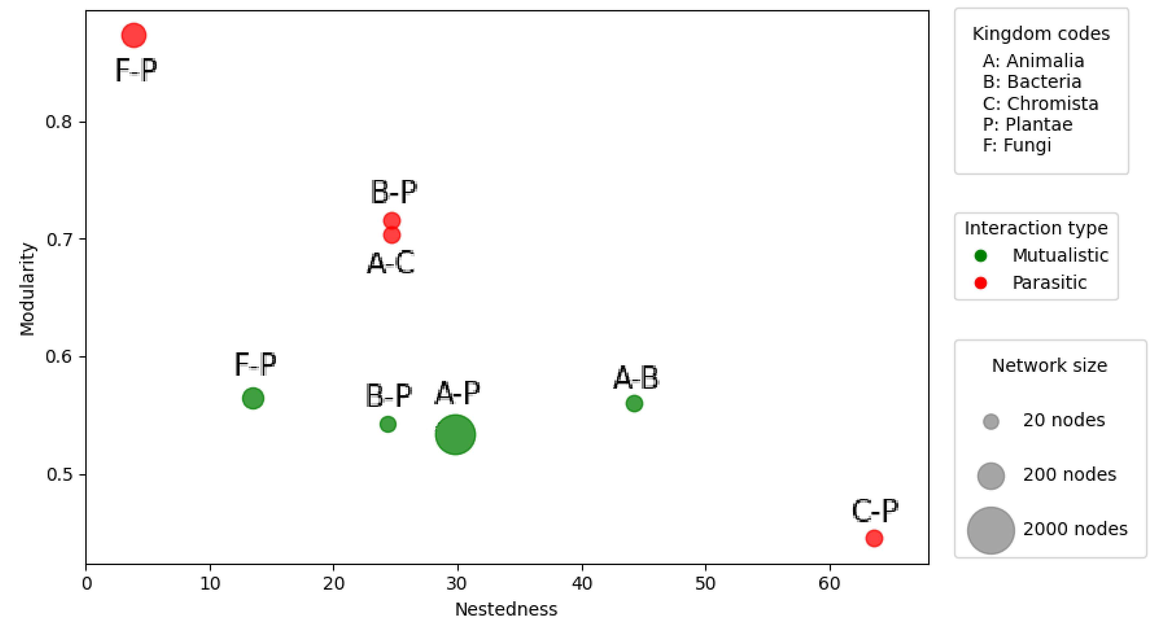
5 Inter-Kingdom Nestedness and Modularity

NODF

$$\frac{100}{k(k-1)/2}\sum_{i<j}\left[\frac{O_{ij}}{min(k_i,k_j)}\cdot\delta(k_i,k_j)\right]$$

Greedy Newman

$$\frac{1}{2m}\sum_{ij}\left[A_{ij}-\frac{k_ik_j}{2m}\right]\delta(c_i,c_j)$$



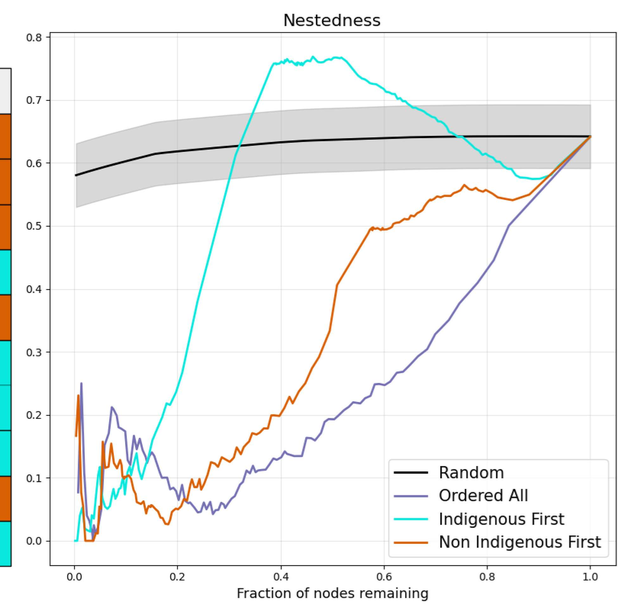
Parasitistic subnetworks tend to be more modular than mutualistic subnetworks

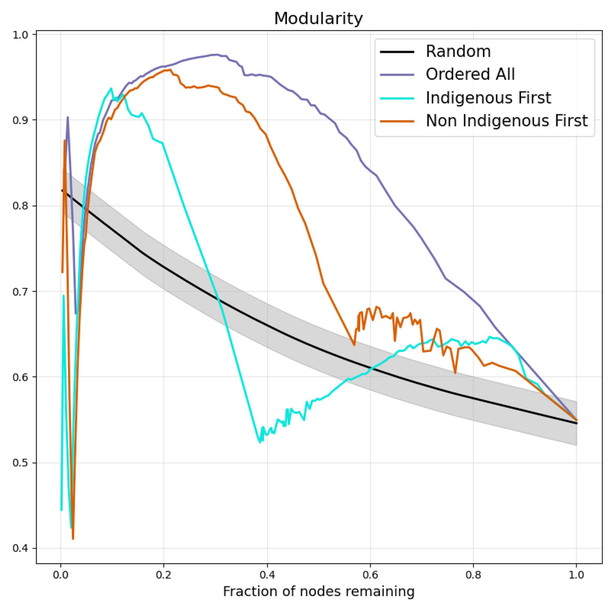
Nestedness expected behaviour wans't clearly observed

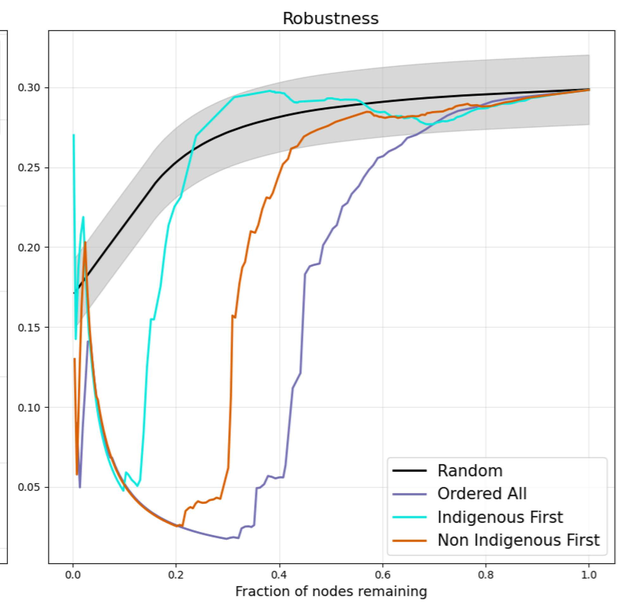
6 Network Resilience

Top 10 species by out-degree

Taxon	Out-degree
Apis mellifera	463
Myzus persicae	341
Macrosiphum euphorbiae	210
Bombus pauloensis	139
Neomyzus circumflexus	129
Colibri coruscans	117
Bombus hortulanus	111
Thygater aethiops	102
Eristalis tenax	98
Tapirus pinchaque	87







Metrics under targeted node removal differ significantly from those under random node removal

Evolution is faster when Non-Indigenous species are removed first

References

[1] A. Del Risco, Á. Montoya, and V. García. "Data synthesis and dynamic visualization converge into a comprehensive biotic interaction network: a case study of the urban and rural areas of Bogotá D.C.". In: *Urban Ecosystems* (2021).

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